UMGS: UAV Multispectral 3D Gaussian Splatting for Spectral-Index Synthesis – Supplementary Material

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SUBMISSION ID: TBD

CCS Concepts: • **Computing methodologies → Image-based rendering**; *Reconstruction.*

Additional Key Words and Phrases: multispectral imaging, gaussian splatting, 3D reconstruction, UAV

ACM Reference Format:
Anonymous Author(s). 2026. UMGS: UAV Multispectral 3D Gaussian Splatting for Spectral-Index Synthesis – Supplementary Material. 1, 1 (May 2026), 3 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Protocol Notes and Scope

This supplement aligns with the current main-paper draft and records additional tables that are useful for reproducibility and reviewer inspection.

The main paper focuses on a 10-scene UAV-focused benchmark: 7 raw UAV captures and 3 aligned aerial multispectral scenes. This supplement separately reports four additional official aligned scenes whose viewpoints are not part of the UAV-focused main scope. The retained-subset diagnostic for raw unaligned input is reported as an applicability check and should not be interpreted as the main raw full-scene comparison protocol.

Depth diagnostics follow the relative reference-depth protocol when the reference source passes validation. Since raw UAV reconstructions are not in a globally calibrated metric coordinate system, all thresholds are relative-depth ratios rather than metric distances. We exclude failed dense-reference attempts from quantitative tables.

Resolution-aligned comparison views. The dagger marker (†) denotes UMGS renders bilinearly resized to the external adapter resolution solely for common-mask comparison. These rows are comparison views, not distinct training variants, and are not used to define the UMGS method.

Grouped split sanitization. For official aligned scenes with inconsistent split and transform metadata, we sanitize splits at the capture-group level against the available transforms. A D/G/R/RE/NIR group is removed only when one or more required members are absent from the transform file, making it unusable as a complete multispectral comparison unit. This is a protocol repair for data completeness, not a performance-based selection.

ACM XXXX-XXXX/2026/5-ART
https://doi.org/10.1145/nnnnnnn.nnnnnnn

Depth reference. Aligned depth diagnostics use sparse point-splat references from available transforms and point support. They are secondary reference-depth evidence, not dense MVS or LiDAR ground truth. The raw `self_m3m` dense MVS mesh-reference attempt failed validation: COLMAP geometric depth maps were empty, normal maps were zero, stereo fusion produced zero fused points, and a manual photometric-depth fallback produced collapsed geometry under CloudCompare inspection. We therefore exclude that dense mesh-reference block from the paper. The geometry claims use only validated support diagnostics and do not claim absolute dense geometry accuracy.

2 Additional Non-UAV Official Scene-Wise External Comparison

Table 1 reports the four additional official aligned scenes that are intentionally kept out of the main UAV-focused tables. Table 2 provides their scene-wise common-mask comparison between UMGS-J† and MS-Splatting. We also include the full seven-scene official aligned average for reproducibility.

3 Depth-Reference Status

We keep the relative-depth evaluator and validation scripts in the released code, but the current paper does not use the failed raw `self_m3m` mesh-reference numbers as evidence. The invalidated artifacts are useful only as an engineering note: sparse COLMAP points were well-formed, but dense MVS did not produce a validated mesh or fused point cloud for that scene. Future dense-reference evaluation should start from a scene whose COLMAP geometric depth, normal maps, stereo fusion, and mesh pass validation before any method comparison is reported.

4 Why Unfrozen-Support Is Not a Product Model

The Unfrozen-support branch is included as a single-band upper-bound ablation. It starts from the same RGB anchor as UMGS, but allows band-specific geometry and opacity updates in Stage 2. This can improve selected per-band render metrics, but it breaks the shared-support contract required by model-level spectral products.

The failure is concrete in our audit logs. On `ms_lake`, Unfrozen-support improves several band PSNR values, but product construction fails with an xyz mismatch between bands:

$$\max |\Delta xyz| = 6.84 \times 10^{-2}.$$

Scratch-MS training diverges further and triggers Gaussian-count mismatch across bands. These variants can still produce independent band images, and one could compute post-hoc 2D ratios after rendering; however, those ratios are not defined over a common 3D Gaussian support and therefore are not used as valid model-level multispectral products in our protocol.

Table 1. Additional official aligned non-UAV scenes, reported separately from the UAV-focused main-paper suite. RGB metrics for UMGS-J use the shared UMGS-I RGB anchor. Spectral metrics average G/R/RE/NIR.

Method	RGB PSNR	RGB SSIM	MS PSNR	MS SSIM	MS LPIPS	Idx. RMSE	Idx. SSIM
UMGS-I	21.67	0.690	25.75	0.763	0.262	0.108	0.710
UMGS-J	21.67	0.690	25.75	0.763	0.262	0.108	0.710

Table 2. Additional non-UAV official aligned scene-level common-mask comparison between UMGS-J† and MS-Splatting. PSNR/SSIM/LPIPS average G/R/RE/NIR; RMSE and SAM are spectral 4-band metrics.

Scene	UMGS-J PSNR	MMS PSNR	UMGS-J SSIM	MMS SSIM	UMGS-J LPIPS	MMS LPIPS	UMGS-J RMSE	MMS RMSE	UMGS-J SAM	MMS SAM
ms_bud	24.77	25.09	0.797	0.781	0.225	0.257	0.0630	0.0601	6.46	7.34
ms_fruit	24.18	23.81	0.698	0.701	0.318	0.306	0.0642	0.0670	8.12	8.66
ms_garden	29.44	27.36	0.874	0.820	0.179	0.283	0.0378	0.0460	3.60	4.40
ms_tree	26.45	24.83	0.812	0.757	0.251	0.325	0.0525	0.0644	5.90	7.07

Table 3. Full seven-scene official aligned common-mask average, included only for reproducibility. Lower is better for LPIPS, 4-band RMSE, and SAM.

Method	PSNR	SSIM	LPIPS	4-band RMSE	SAM°
UMGS-J†	26.55	0.820	0.231	0.0544	5.75
MS-Splatting	26.03	0.803	0.272	0.0571	6.23

5 UMGS-J Initialization Ablation (UMGS-J vs UMGS-JR)

The main paper keeps UMGS-JR out of the main tables and reports UMGS-J as the representative joint branch. This section records the initialization ablation that motivated that decision. Both variants use the same geometry-locked joint multispectral training design and export per-band evaluation views from a unified checkpoint. They differ only in appearance-bank initialization.

UMGS-J initializes each spectral bank from the reset-appearance state. UMGS-JR initializes each spectral bank from the RGB-tied carrier.

Table 4 compares these variants on the two joint-branch pilot scenes used during method screening. UMGS-JR does not improve the branch: it degrades image/index quality on both pilot scenes.

6 Takeaway

The supplementary tables support three practical points used by the paper: (1) additional official aligned scenes remain available for reproducibility but are outside the main UAV-focused scope, (2) depth-reference evidence is reported only when the reference source passes validation, and (3) UMGS-JR is an initialization ablation rather than a promoted branch.

Table 4. UMGS-J and UMGS-JR quality/cost comparison from pilot runs. Band metrics are G/R/RE/NIR means. Index metrics are NDVI/GNDVI/NDRE means.

Scene	Variant	Band SSIM ↑	Band PSNR ↑	Band LPIPS ↓	Index MAE ↓	Index RMSE ↓	Index PSNR ↑	Index SSIM ↑	Time (s) ↓
`self_m3m`	UMGS-J	0.7657	25.330	0.2113	0.0607	0.0863	27.701	0.7943	8129
`self_m3m`	UMGS-JR	0.7434	24.966	0.2061	0.0623	0.0880	27.519	0.7854	7339
Vineyard A	UMGS-J	0.8804	28.690	0.1092	0.0339	0.0513	32.169	0.8948	5805
Vineyard A	UMGS-JR	0.8590	27.847	0.1266	0.0375	0.0574	31.326	0.8800	6730